

House Price Prediction using Feature Engineering & Model Comparison

Objective

The objective of this project is to enhance the House Price Prediction model developed in Task 1 by introducing feature engineering techniques and comparing multiple regression algorithms. The project demonstrates how different machine learning models perform on the same dataset and identifies the most suitable model.

Dataset

California Housing Dataset

- 20,640 records
- 8 features
- Target: Median House Value

Feature Engineering

Feature Scaling was performed using StandardScaler.

Scaling ensures all features are on the same numerical scale, improving training stability and model performance.

Models Used

- Linear Regression
- Ridge Regression

- Decision Tree Regressor

Evaluation Metrics

Three metrics were used:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Include:

- Model Comparison Table screenshot
- R^2 Comparison Graph
- Actual vs Predicted Graph

Best Model

Mention whichever model got the highest R^2 score in the notebook.

Example:

Decision Tree Regressor achieved the highest R^2 score and lowest RMSE, making it the best-performing model for this dataset.

(or Linear Regression/Ridge Regression if that was the result)

Conclusion

This project successfully demonstrated the use of feature engineering, feature scaling, model optimization, and performance comparison using

multiple regression algorithms.

The workflow follows industry-standard machine learning practices and highlights the importance of selecting the most suitable model based on evaluation metrics.